

Sourabh Porwal

Functional Safety Engineer · Safety-Critical Systems

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Functional Safety / Systems Engineering / Autonomous & Unmanned Systems

PROFILE

Seven years in functional safety for automotive and autonomous driving. Currently own functional safety at Vay, whose fleet of around 200 remotely driven vehicles operates on public roads in Las Vegas with no one in the vehicle. No regulation existed for teledriving, so the safety framework was derived from first principles and the in-house safety function built from scratch, through to a positive independent assessment by TÜV SÜD and Vay's first-in-Europe approval to operate without a safety driver. Deep in ISO 26262 and ISO 21448 across the full lifecycle: hazard analysis, safety cases, requirements, verification. Now leading the safety work as AI enters the driving task itself.

PROFESSIONAL EXPERIENCE

Senior Functional Safety Engineer, Vay Technologies GmbH, Berlin Sep 2023 – Present

- Own functional safety for Vay's remote-driving system, driving ISO 26262 compliance across the systems, hardware, and software teams from concept through real-world operation on public roads.
- Own and author the safety case to UL 4600, and define the functional and technical safety architecture, including system-level redundancies, for a fleet of ~200 remotely driven vehicles in Las Vegas.
- Owned end-to-end development of the fail-safe fallback behaviour (Minimum Risk Manoeuvre) from concept through functional safety, including quantitative frequency targets for commercial launch, applying ISO 26262, UNECE R 157, and ISO 4804.
- Derive tolerable failure-rate thresholds for each fallback-manoevre type from probabilistic risk analysis, with an operational stop-fleet trigger if the thresholds are exceeded.
- Led the functional-safety side of the independent TÜV SÜD assessment of the safety concept and risk analyses, contributing to Vay's first-in-Europe approval to operate on public roads without a safety driver.
- Act as safety lead for integrating Vay's remote-driving stack into Kodiak's autonomous-driving product as a fallback layer, working directly with their safety and engineering teams on architecture, integration, and safety deliveries.
- Lead the safety work for bringing AI into safety-critical driving functions.

Functional Safety Engineer II / I, Vay Technologies GmbH, Berlin Jul 2021 – Aug 2023

- Helped establish Vay's in-house functional-safety function and processes from scratch, deriving the safety framework from first principles in the absence of any teledriving regulation.
- Delivered complete ISO 26262 work-product sets across release cycles: HARA, safety goals with ASIL classification, FMEA, FMEDA, FTA, safety concepts, and safety requirements.
- Translated ambiguous, first-of-their-kind safety goals into structured requirements, verification plans from system to vehicle level, and explicit closure criteria, culminating in a positive external safety assessment.
- Defined the interface between the safety controller and the vehicle over CAN and LIN, characterised undocumented vehicle signals, and authored the resulting database changes, from failure-mode analysis through to deployed fixes.
- Implemented safety requirements management and traceability in JAMA.

Functional Safety Engineer / Manager, BERNIS Engineers GmbH, Gilching Jan 2020 – Jun 2021

- Prepared safety concepts and analyses (FSC, TSC, HARA, FTA) for OEM customers and worked directly with their architects on ASIL D ADAS features.
- As functional safety manager on certain projects, owned the safety plan, oversaw supplier safety activities, and provided functional-safety releases for the OEM's internal development milestones.

Functional Safety Engineer, AKKA Technologies, Ingolstadt Jun 2019 – Dec 2019

- Supported ISO 26262 development for an ASIL B system and managed vehicle configuration data.

Junior Executive, Fleetguard Filters Pvt. Ltd., Pune, India Aug 2013 – Sep 2015

- Managed the production part approval process (PPAP) across manufacturing suppliers, including process FMEA; reduced component cost by ~25% through supplier negotiation.

SKILLS

Standards: ISO 26262 (full lifecycle), ISO 21448 (SOTIF), UNECE R 157, ISO 4804, UL 4600 (safety case), FMVSS, ISO/IEC TS 5083; IEC 61508 principles applied through ISO 26262

Safety analysis: HARA, FMEA, FMEDA, FTA, functional and technical safety concepts, safety cases, quantitative risk analysis

Systems engineering: Requirements engineering and traceability, V-model development, safety architecture, verification and validation at system, HiL, and vehicle level, CAN/LIN interface definition

Regulatory: Translating regulation into engineering deliverables: teledriving ordinance (StVFernLV) scoping, permit documentation, ODD qualification, FMVSS compliance

Tools: IBM DOORS, JAMA, MATLAB/Simulink, Isograph, APIS IQ

Programming: C/C++, Python (general working knowledge)

EDUCATION & CERTIFICATION

M.Sc. Mechanical Engineering, Rhine-Waal University of Applied Sciences, Kleve Oct 2015 – Apr 2019

B.E. Mechanical Engineering, Visvesvaraya Technological University, Belgaum Aug 2009 – Apr 2013

CertX ISO 26262:2018 Functional Safety Red Belt Mar 2021

PERSONAL DETAILS

Nationality: German (eligible for German security clearance, ZÜP)

Mobility: Open to relocation (Berlin to Munich)

Languages: English (fluent), German (B1, working proficiency), Hindi (native), Marathi (native)